

Broker Routing Decisions in Limit Order Markets

David A. Cimon*

July 23, 2018

ABSTRACT

Many investors rely on brokers to route their orders to exchanges. Exchanges charge fees to the broker who routes the order, rather than to the investor. Brokers have an incentive to route based on the fees, instead of the execution quality experienced by their clients. This conflict of interest reduces welfare, as investor limit orders are sent to exchanges with the lowest fees rather than the best market quality. By limiting exchange fees or passing fees to investors, a regulator can both remove the conflict of interest and improve welfare.

JEL Classification: G24, G28

Keywords: brokers, market microstructure, market quality, market fragmentation, make/take fees

*Bank of Canada, Financial Markets Department.

E-mail: dcimon@bank-banque-canada.ca. Tel: +1 613-782-7612

I would like to thank Andreas Park for his invaluable guidance. I would like to thank Jason Allen, Robert Battalio, Michael Brolley, Jean-Sébastien Fontaine, Corey Garriott, Lawrence Glosten, Charles Jones, Jordi Mondria, Cheng Zheng, and staff from the Ontario Securities Commission and Investment Industry Regulatory Organization of Canada for their helpful discussion. I would also like to thank seminar participants at the University of Toronto, Copenhagen Business School, HEC Montréal, Université Laval, and the Bank of Canada, as well as conference participants at the 2016 NFA, the 2015 SFS Cavalcade, the 2015 ERIC Doctoral Consortium, the 2015 TADC, and the 2015 CEA Annual Conference for their helpful feedback. The views expressed in this paper are those of the author and not necessarily those of the Bank of Canada. All errors are my own.

Many investors do not access markets directly; instead, they rely on a broker to route their orders to one of many exchanges. Because they pay a commission, investors may believe that their broker will represent their best interests. However, most exchange operators set fee structures to incentivize brokers to route orders to their exchanges. One such fee structure is known as “maker-taker” pricing, whereby the exchange provides rebates for liquidity-supplying orders. Concerns that fee structures affect order routing by brokers were raised as early as 2000 by the SEC, and have been discussed more recently in 2014 by a U.S. Senate Committee, and in 2017 by a U.S. Treasury Report.¹ Recent empirical evidence suggests that brokers’ order routing behavior is influenced by these fees (Battalio, Corwin, and Jennings 2016).

I study the impact of exchange fees on brokers’ routing incentives by constructing a model of trading on the limit-order book. The main result is that exchange fees can create an inefficiency: brokers choose to route investors’ limit orders to exchanges with poor market quality but high rebates. As a result, investors realize fewer gains from trade and face more informed traders. In the model, losses to investors are driven by the fact that they do not fully observe market conditions. Specifically, investors do not observe the state of the limit-order book directly. This information disadvantage creates an agency conflict between investors and their brokers.

The model generates new testable predictions as well as potential policy prescriptions. First, exchanges with lower liquidity-taking fees will have higher execution probabilities and

¹See SEC “Proposed Rule: Disclosure of Order Routing and Execution Practices” <http://www.sec.gov/rules/proposed/34-43084.htm>.

The hearings, entitled “Conflicts of Interest, Investor Loss of Confidence, and High Speed Trading in U.S. Stock Markets” are available: <http://www.hsgac.senate.gov/subcommittees/investigations/hearings/conflicts-of-interest-investor-loss-of-confidence-and-high-speed-trading-in-us-stock-markets>. The U.S. Treasury’s Second Report On The Administration’s Core Principles Of Financial Regulation is available here: <https://www.treasury.gov/press-center/press-releases/Pages/sm0173.aspx>.

lower adverse selection measures. When prices are constrained by tick sizes, traders using market orders prefer these venues, increasing execution quality measures for limit orders. Second, when brokers charge higher commissions, they will be willing to send some limit orders to these low liquidity-taking fee venues, whereas when they charge lower commissions they will send all limit orders to venues with low liquidity-making fees. Finally, the model predicts that brokers will have an even stronger preference for venues with liquidity-making rebates when there is a higher probability of informed trading in the market.

The model suggests two regulatory actions which may be effective in removing the conflict of interest studied in this paper. First, regulators can place limits on fees charged or rebates paid by exchanges, which reduces the conflict of interest. The conflict can be removed entirely if regulators require identical liquidity-supply fees across venues. However, the impact on welfare depends on the fee structures of incumbent exchanges and on the fee selected by the regulator. Second, regulators can limit welfare losses by mandating that brokers pass through all exchange fees. However, unless investors are able to signal complex state-contingent routing priorities, passing through exchange fees does not necessarily achieve a first-best welfare outcome.

There are three essential features in the model which, taken together, differentiate this paper from the prior literature. First, a single security trades at two exchanges with a fixed tick size, differentiated by trading fees. The use of fixed tick sizes creates pricing frictions, similar to those present in real-world trading, where agents are unable to set continuous prices. As in practice, the exchanges charge separate fees for providing liquidity through limit orders (referred to as “making liquidity”) and demanding liquidity through market orders (referred to as “taking liquidity”). By assumption, the fees are smaller than the tick size, negating the situation in Colliard and Foucault (2012) where investors adjust prices to

compensate for fees. This assumption matches the reality of most equities trading, where the tick size is \$0.01 but fees are less than \$0.003 per share.²

Second, to access these exchanges, investors must transact through a broker. These brokers are allowed to offer multiple commissions, which dictate their routing strategy. From a theoretical angle, allowing multiple commissions represents a conservative estimate of losses to investors. Were brokers restricted to charging any single commission, welfare losses to investors would be greater, while other results would remain similar.³ In practice, many brokers offer different commission structures. Some common pricing structures include: pricing per-share, flat commissions, and tiered-pricing based on order size. This diversity in fee structure is especially common following the proliferation of low-cost Internet brokerages, whose commissions are often dramatically lower than full-service brokerages.⁴

Finally, the key friction is that investors are unable to view the state of the limit-order book when their broker routes their order. In practice, even investors with real-time pricing data on their broker's trading platform are often unable to view the exact state of the order book at multiple exchanges when their order is routed. Due to the speed of modern markets, conditions can very quickly change between the time an investor submits an order, and it is ultimately routed by their broker. To view the real-time state of the limit-order book, investors would require access to prohibitively expensive smart-order routers, real-time data feeds and colocation services from multiple exchanges. In the model, brokers do have access to the technologies that allow them to view the limit-order book's current state. This speed friction creates a conflict between investors and their brokers: investors cannot

²Regulated under SEC Rule 610.

³An example of one such commission is the one generated in Brolley and Malinova (2013).

⁴Existing research on broker competition has argued that commissions from full-service brokerages have dropped dramatically from upwards of \$75 USD per 100-share trade in the 1990s (Bakos, Lucas Jr, Oh, Simon, Viswanathan, and Weber 2005). Currently, some online brokerages (such as Interactive Brokers) charge as little as \$1 per 100 share trade.

determine the exact conditions under which their broker chose an exchange, while brokers earn informational rents from investors willing to pay for superior market quality.

A. Existing Literature

In this paper, exchanges charge fees based on supplying and demanding liquidity. Thus, this paper contributes to the growing literature on “maker-taker” pricing. Empirically, several papers have found that exchanges with a maker-taker structure often have a better spread posted (Malinova and Park 2015, Anand, McCormick, and Serban 2013). However, exchanges with either a maker-taker structure or a higher taker fee have also been found to have a higher concentration of informed trading (Yim and Brzezinski 2012, Anand, McCormick, and Serban 2013). My paper supports this latter result, as uninformed traders using market orders prefer exchanges with the lowest fees for taking liquidity.

Existing theoretical work on “maker-taking” pricing focuses on the impact of fees on investors who submit orders directly to exchanges. Colliard and Foucault (2012) find that only the total fee has an effect on investor outcomes, and that the breakdown between maker and taker fees has no effect on the division of gains from trade. Similarly, Maglaras, Moallemi, and Zheng (2015) focus on investors who queue orders and optimally weigh the trade-offs between exchange fees and market quality across venues. Brolley and Malinova (2013) find that market orders sent to maker-taker exchanges are subsidized by investors who submit limit orders, when investors pay a flat commission. My paper extends these results by showing how the breakdown between maker and taker fees plays a role when brokers, rather than investors, route orders across multiple exchanges.

This paper provides theoretical support for existing empirical work on broker-routing behaviour. Battalio, Corwin, and Jennings (2016) find that brokers often make routing

decisions based on liquidity rebates and that investors face higher adverse selection costs at exchanges with higher liquidity rebates. Bakos, Lucas Jr, Oh, Simon, Viswanathan, and Weber (2005), find that discount brokerages are more likely than traditional brokerages to route orders to third-party dealers as opposed to the primary exchange. My model supports both these results: brokers route orders based on fees, and brokers display different routing behaviour based on their commissions. Boehmer, Jennings, and Wei (2007) find that when access to statistics on execution quality increases, competition for order flow among broker-dealers drives orders to venues with low effective spreads and quicker average fills. My model presents a best-case scenario in which investors correctly anticipate execution quality at each venue: were investors to have less access to this information, welfare losses would only be greater.

Brokers may have options other than exchanges for order execution. In some circumstances, brokers may simply transact orders against their own inventory, rather than exposing them to the market. Existing work describes the incentives for dealers when handling client order flow against their own inventory (Stoll 1978, Amihud and Mendelson 1980, Ho and Stoll 1981). Alternatively, the literature on payment for order flow focuses on brokers who receive payments for marketable flow by off-exchange market makers (Battalio and Holden 2001, Parlour and Rajan 2003). In both cases, existing work on off-exchange execution focuses on marketable order flow. I expand on these papers by studying broker incentives when handling limit orders, where uncertainty in order execution plays a key role.

I. Model

A. Trading Environment

There are three periods, $t \in \{0, 1, 2\}$. There is a single security with a value of $v = 0$ at $t = 0$ and V at $t = 2$. V is unknown in $t = 0$ and $t = 1$. At $t = 2$, one of two outcomes occurs. With probability δ , an innovation occurs and $V = -\sigma$. Alternatively, no change occurs and $V = 0$.⁵

There are two exchanges with identical bid prices given by $-\Delta$, where $\Delta > 0$. Exchanges differ based on the fees they charge. Each Exchange i charges exogenous fees M_i and T_i to traders for orders making liquidity (limit orders) and taking liquidity (market orders), respectively.⁶

As in practice, each exchange operates on a time-priority system. A limit order that arrives at Exchange i in $t = 0$ receives priority over a limit order that arrives at Exchange i in $t = 1$. However, time priority does not hold across exchanges. A limit order that arrives at Exchange i in $t = 0$ does not receive priority over a limit order at Exchange j in $t = 1$.

I make three simplifying assumptions on the exchange parameter set.

Exchange Fee Assumptions

Assumption E1: The total cost per trade is set to zero, $M_i + T_i = 0$.

Assumption E2: The taker fee is higher at Exchange 2, $T_2 > T_1$.

Assumption E3: The fees and tick size are smaller than the innovation, $\Delta + T_i < \sigma$.

⁵I focus on negative innovations and limit orders to buy the security. Results for positive innovations and limit orders to sell are symmetric.

⁶I do not consider identical exchanges or exchanges with endogenous exchange fees. In practice, though many exchanges offer maker-taker pricing, the exact fees charged by exchanges vary greatly. One example of work on endogenous exchange fees is Chao, Yao, and Ye (2015).

Assumption E1 is consistent with zero-profit exchanges that have zero total cost per trade. This assumption is not critical for the results; relaxing it would allow the exchanges to extract profits, but does not change the incentives of agents. Assumption E2 simplifies exposition. Assumption E3 removes the economically uninteresting cases where one or both exchanges receive no orders from informed traders when an innovation in the asset value occurs. Were the opposite true, standing limit orders would face no adverse selection risk at this exchange.

B. Market Participants and Timing

The market participants are: (i) market makers, (ii) limit-order investors, (iii) brokers, (iv) informed traders, and (v) liquidity traders. Market makers post orders at $t = 0$, the market-making period. Limit-order investors may post orders in $t = 1$, the liquidity-supply period. Market orders are posted in $t = 2$, the liquidity-demand period. The timing of the model is illustrated in Figure 1.

[Figure 1 about here.]

i. Market Makers. An exogenous market maker automatically provides liquidity on the exchanges. The market maker arrives at the beginning of $t = 0$, and may place a single order to buy at one of the exchanges, which remains available until the end of $t = 2$. The location of the market maker's order (if any) is referred to as the state (S) of the market. With probability p_1 , the market maker places his order at Exchange 1 ($S = 1$), while with probability p_2 , he places his order at Exchange 2 ($S = 2$). Otherwise, with probability $1 - p_1 - p_2$, no order is placed ($S = 0$).

ii. Limit-Order Investors. One limit-order investor arrives at the beginning of period $t = 1$. She has an exogenous desire to buy one unit of the security. The investor has a

private value y drawn from a uniform distribution $U[0, y]$, as in Parlour (1998) and Foucault (1999).⁷

To conduct a trade, the investor places her order with a broker. She views the menu of commissions offered by the brokers and selects a commission, c_n , which she pays if the order executes. Orders are at the price $-\Delta$ and remain at the exchange until the end of $t = 2$. The investor is at an information disadvantage compared to her broker: while she observes her order's routing she does not view the state of the limit-order book when her order is routed. The investor has knowledge of the broker's profit function and believes that the broker will route to the exchange which is most profitable, given the commission, in each state.

A limit-order investor's utility function for a limit-order buy, which is subsequently routed to Exchange i in State S , is given by:

$$U(S) = \theta_i(S) (E[V|ex_i(S)] + y + \Delta - c_n), \quad (1)$$

where $\theta_i(S)$ is the probability that her order will execute at Exchange i in State S and $E[V|ex_i(S)]$ is the expected value of the asset, for the limit-order investor, conditional on execution at Exchange i in State S .

iii. Brokers. There exist a group (size $N > 2$) of brokers who are risk-neutral, uninformed about the innovation in the asset value, and are profit maximizing. Brokers receive orders from investors, view the state of the limit-order books and route their client's order to one of the exchanges. Brokers profit from the difference between the commissions they charge their clients and the costs they pay to route orders (M_i for limit orders at Exchange i). If no execution occurs, the brokers incur no costs and receive no commission.

Each broker is able to set a menu of commissions C , composed of individual commissions, indexed c_n . Limit-order investors select one of the commissions when sending an order to a

⁷A continuous distribution of private values is useful to derive welfare implications.

broker. This mirrors the real-world structure of the brokerage market, where brokers often quote multiple commissions for different tiers of service.⁸

For orders routed to Exchange i in State S at Commission c_n , the broker's profit is given by:

$$\pi = \theta_i(S)[c_n - M_i]. \quad (2)$$

While the broker's routing decision is observable by investors, the current state of the limit order book is not. Limit-order investors are aware of this fact and, thus, brokers are subject to an incentive compatibility constraint. In any given state, a broker receiving a Commission c_n will route the order to Exchange i , if it is more profitable than Exchange j . Mathematically, this is expressed as:

$$\theta_i(S)[c_n - M_i] \geq \theta_j(S)[c_n - M_j]. \quad (3)$$

iv. Informed Traders. With probability δ , an innovation occurs. If so, an informed trader arrives at the market and is the only agent to view the innovation value.⁹ The informed trader views the state of the limit order book, as well as any order submitted by the limit-order investor.

The trader immediately submits market orders for all available orders at both markets. The informed trader's order is self-routed, and they pay all fees associated with taking liquidity (T_i). However, this routing does not play a role in equilibrium, as the informed trader simultaneously submits orders for all available liquidity. Thus, all exchanges with available liquidity receive informed orders.

⁸It should be noted that in equilibrium, this structure improves the outcome for investors. Brokers who charged any single commission would only worsen the welfare results found later in this paper.

⁹This trader presents an adverse selection risk to limit-order traders who have already placed orders, similar to Glosten and Milgrom (1985), Easley and O'Hara (1987), Glosten (1994) and others.

v. Liquidity Traders. With probability $1 - \delta$, no innovation occurs. If so, a liquidity-demanding investor arrives with a quantity signal, distributed evenly over $Q_2 \in \{1, 2\}$.¹⁰ She views the state of the limit order book, as well as any order submitted by the limit-order investor. The liquidity trader then submits market order sells for the total amount of her desired quantity, which she routes to the exchanges herself.¹¹ Liquidity investors pay all fees associated with taking liquidity (T_i) and thus, route to venues with lower liquidity taking fees when possible. Liquidity investors who wish to trade, but cannot because of insufficient liquidity, pay a cost $k > \Delta + \max\{T_1, T_2\}$ for each desired order which they are unable execute.

II. Equilibrium

In this section, I present the model equilibrium. An equilibrium is a solution to the liquidity investor's market-order routing problem, the brokers' limit-order routing problem and the limit-order investor's utility maximization problem. I search for a weak perfect Bayesian equilibrium, using a guess and verify approach.

A. Market Order Routing

First, I detail the liquidity trader's decisions in $t = 2$. Liquidity traders route one or both orders to Exchange 1, if liquidity is available, since liquidity taking fees are lower under Assumption E2 ($T_1 < T_2$). Only if there is insufficient liquidity at Exchange 1, will orders

¹⁰The inclusion of these traders removes the no-trade equilibrium detailed in Milgrom and Stokey (1982).

¹¹Unlike limit-order investors, liquidity traders would face no conflict of interest with brokers in this model. Limit orders are at identical prices and contain no information in this model. Thus, factors other than exchange fees play no role for liquidity traders. Were brokers to route market orders, their preferences would be identical to those of the liquidity traders, with both selecting any available liquidity, with a preference for the lowest taker fee.

be sent to Exchange 2. This information determines the expected execution probability and asset values for limit-order investors, depending on the state of the order book.

While Exchange 1 always receives priority for market orders, a limit-order investor's order may not be the first to arrive. When the market maker places its order at Exchange 1 ($S = 1$), it receives priority over the limit-order investor. On the other hand, when a market maker places its order at Exchange 2 ($S = 2$) or does not place an order ($S = 0$), limit orders from the investor at Exchange 1 receive first priority for market orders. I combine this with the fact that informed traders send orders equally to all exchanges, to determine the equilibrium expected asset value and execution probability at Exchange 1. The expected value of the asset at Exchange 1 is given by:

$$E[V|ex_1(S)] = -\delta \cdot \sigma \quad \text{if } S \in \{0, 2\}, \quad (4)$$

$$E[V|ex_1(S)] = \frac{-2\delta \cdot \sigma}{(1 + \delta)} \quad \text{if } S = 1. \quad (5)$$

The probability of execution at Exchange 1 is given by:

$$\theta_1(S) = 1 \quad \text{if } S \in \{0, 2\}, \quad (6)$$

$$\theta_1(S) = \frac{(1 + \delta)}{2} \quad \text{if } S = 1. \quad (7)$$

Similarly, the state of the limit-order book determines the equilibrium expected asset value and execution probability at Exchange 2. If market makers post at either exchange ($S \in \{1, 2\}$), the limit-order investor is behind them in priority for market orders if their order is sent to Exchange 2. A limit-order investor only receives first priority at Exchange 2 if no market maker posts ($S = 0$). Thus, the expected value of the asset at Exchange 2 is

given by:

$$E[V|ex_2(S)] = -\delta \cdot \sigma \quad \text{if } S = 0, \quad (8)$$

$$E[V|ex_2(S)] = \frac{-2\delta \cdot \sigma}{(1 + \delta)} \quad \text{if } S \in \{1, 2\}. \quad (9)$$

The probability of execution at Exchange 2 is given by:

$$\theta_2(S) = 1 \quad \text{if } S = 0, \quad (10)$$

$$\theta_2(S) = \frac{(1 + \delta)}{2} \quad \text{if } S \in \{1, 2\}. \quad (11)$$

Lemma 1 (States and Market Quality):

The probability of execution and expected value of the asset that a limit-order investor expects at each exchange, are different only if market makers post at Exchange 2 ($S = 2$). Otherwise, the limit-order investor expects identical market conditions at both exchanges.

Lemma 1 defines the scope of assets where a conflict of interest occurs. The choice of exchange matters, and thus conflicts of interest can occur, only if market makers place orders at Exchange 2.¹² Exchange 2 has the lower liquidity-making fee, and thus, can be seen to correspond to a conventional “maker-taker” venue. While market making is exogenous in this model, in practice, maker-taker venues receive high-volumes of orders from liquidity providers. Thus, the conflict of interest is present in these cases where market makers post at the best price on maker-taker venues.

¹²This would also be true, were market maker able to post at both, rather than only a single venue.

B. Limit Order Routing

Given the expected routing of market orders in $t = 2$, the broker's problem in $t = 1$ is to set commissions and route orders. First, each broker is able to announce a menu of commissions C . Brokers then receive order flow from investors and route orders based on the commission chosen by the investor. In each state, the broker selects the exchange which is most profitable for himself, given the commission selected.

A valid equilibrium in commissions and routing strategies is one in which: i) all commissions are incentive compatible given their expected routing strategy, and ii) it is not profitable for any single broker to deviate, either by changing one of their commissions or by offering a new commission. I conjecture that there exists an equilibrium, in which all N brokers offer an identical menu consisting of two commissions: a higher commission, denoted c_H and a lower commission, denoted c_L .

First, I conjecture that the Commission c_L is a zero-profit commission in which brokers route to Exchange 2 in all states. This commission is defined:

$$c_L = M_2. \quad (12)$$

Second, I conjecture that the Commission c_H is a commission in which brokers route to Exchange 2 in states $S = 0, 1$ and to Exchange 1 in State $S = 1$. I propose that this commission is defined as the lowest commission such that it is incentive compatible for the broker to route to Exchange 1 in State $S = 2$. This is defined:

$$\theta_1(S = 2)[c_H - M_1] \geq \theta_2(S = 2)[c_H - M_2] \quad (13)$$

By inserting the values for $\theta_1(S = 2)$ and $\theta_2(S = 2)$, manipulation of Equation 13 results in:

$$c_H = \frac{2M_1 - (1 + \delta)M_2}{1 - \delta}. \quad (14)$$

The equilibrium commission in which brokers occasionally route to Exchange 1 is strictly higher than the commission under which brokers do not ($c_H > c_L$). This relationship occurs, both because maker fees are higher at Exchange 1, and because brokers must earn positive profits at Exchange 1 for this strategy to be incentive compatible. These positive profits represent an informational rent, which brokers extract from limit order investors. An equilibrium in which that lower priced brokers route orders exclusively to Exchange 2 is consistent with the empirical findings of Battalio, Corwin, and Jennings (2016), who find that popular retail brokers route almost all non-marketable order flow to the venue with the highest maker rebate.

C. Limit Order Submission

Finally, I solve for limit-order investor's solutions in $t = 1$, given the brokers' commission schedules and the expected routing of market orders. The solution to the limit-order investor's problem is a decision of both whether to submit an order and which commission to pay, given her private signal y . The limit-order investor does not view the state, and must make her selection based on her expectations of broker behaviour.

The investor's decision takes the form threshold private values y_H and y_L . These thresholds are derived from the limit-order investor's profits under each routing strategy. Given the equilibrium actions of the brokers and market-order traders, the expected utilities of an

investor who selects commissions c_H and c_L are defined:

$$U_H(y) = p_1 \frac{(1+\delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1+\delta)} + y - c_H + \Delta \right] + p_2 [-\delta \cdot \sigma + y - c_H + \Delta] \\ + (1 - p_1 - p_2) [-\delta \cdot \sigma + y - c_H + \Delta], \quad (15)$$

$$U_L(y) = p_1 \frac{(1+\delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1+\delta)} + y - c_L + \Delta \right] + p_2 \frac{(1+\delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1+\delta)} + y - c_L + \Delta \right] \\ + (1 - p_1 - p_2) [-\delta \cdot \sigma + y - c_L + \Delta]. \quad (16)$$

By inspection, Equation 15 is greater than Equation 16 for y sufficiently high. Thus, the threshold y_H is the value of y such that for all $y \geq y_H$ the limit-order investor prefers to select Commission c_H . Limit-order investors with a private value $y < y_H$ will choose to either pay the Commission c_L , or not trade. Any investor who has a private value y such that Equation 16 is positive is better off trading at Commission c_L than not trading. Thus, the value y_L is the value of y such that investors with a value $y_H > y \geq y_L$ will choose to trade at Commission c_L . Formally, these values y_H and y_L are the lowest values of y that satisfy the following two equations, respectively:

$$U_H(y) \geq \max\{U_L(y), 0\}, \quad (17)$$

$$U_L(y) \geq 0. \quad (18)$$

Proposition 1 (Existence of an Equilibrium):

There exists a weak perfect Bayesian equilibrium, depicted in Figure 2, such that:

- (i) N brokers offer identical commission menus, consisting of two commissions c_L and c_H , defined by Equations 12 and 14.*
- (ii) Investors select to submit orders at commissions c_L and c_H , or to not trade, based on the thresholds y_H and y_L defined by Equations 17 and 18.*

[Figure 2 about here.]

III. Results

A. Market Conditions and Outcomes

In this section, I present results related to market conditions and welfare. First, I discuss the effects of market order routing on both execution probability and expected asset value, on each exchange. Second, I discuss the utility and welfare effects of broker behavior.

Market conditions at the two exchanges result directly from the preferential routing of market orders to the venue with the lower taker fee. Since market orders will only be sent to Exchange 2 when liquidity at Exchange 1 is exhausted, limit orders posted at Exchange 1 necessarily have a higher fill rate. Further, because informed orders are always for all available liquidity, orders at both exchanges receive all informed orders, but different quantities of uninformed orders.

Proposition 2 (Market Order Routing and Market Quality):

- (i) *The execution probability for limit orders at Exchange 1, the low taker fee exchange, is greater than or equal to that at Exchange 2 in all states ($\theta_1(S) \geq \theta_2(S) \ \forall S$).*
- (ii) *For limit orders, the expected value of the security conditional on execution, is weakly higher if it is routed to Exchange 1, than if it is routed to Exchange 2 in all states ($E[V|ex_1(S)] \geq E[V|ex_2(S)] \ \forall S$).*

The first statement in Proposition 2 deals with the total number of orders received by each exchange. Both exchanges receive every informed order when liquidity is available, while Exchange 1 receives more uninformed orders, increasing its execution probability. The

second statement deals with the relative probability of receiving informed orders. As above, the each exchange receives all informed orders, while Exchange 1 receives more uninformed orders. Thus, Exchange 1 receives a lower proportion of informed orders, relative to its total number of orders, increasing the expected value of the asset for limit order buyers. This result supports the conclusions of empirical work from Anand, McCormick, and Serban (2013) and Battalio, Corwin, and Jennings (2016), who find that exchanges with a maker-taker structure have a greater concentration of informed order flow.

Testable Implication 1 (Execution Probability and the Probability of Informed Trading):

Of any two exchanges, the one with lower liquidity-taking fees will have a higher execution probability and lower concentration of informed trading.

The first testable implication of this model concerns market quality. When two markets post the same spread, the one with lower liquidity-taking fees should have better market quality, in terms of higher execution probabilities and lower concentrations of informed trading. The same is true for a venue which decreases its liquidity-taking fees below that of a competing venue. Following a decrease in liquidity-taking fees, a venue should see improved market quality, while the venue whose fees are now higher should see market quality deteriorate.

I compare the equilibrium in Proposition 1 to a benchmark case (“the Benchmark”) without brokers. In the Benchmark, limit-order investors access markets directly: they are able to view the state of the order book, route their orders manually and pay all associated trading fees. I represent variables from the Benchmark through the addition of the subscript B . While there are no brokers in the Benchmark, investors continue to sort based on their private value y , as in Proposition 1. During State $S = 2$, limit-order investors with $y \geq y_{H,B}$

select Exchange 1, while limit-order investors with $y_{H,B} > y \geq y_{L,B}$ select Exchange 2. All limit-order investors continue to select Exchange 2 during State $S = 0$ and State $S = 1$.¹³

Proposition 3 (Welfare and Transfers):

Compared to the Benchmark case without brokers, when investors must access markets through brokers:

- (i) These brokers earn positive profits in expectation, and limit-order investors have lower utility in expectation.*
- (ii) Overall welfare declines.*

The combination of different conditions at the two exchanges and the conflict of interest faced by brokers, creates the key result of this paper: the inefficiency of trading fees. The difference in fees at exchanges provides the driving force behind the principal-agent problem between brokers and their clients. When routing limit orders, brokers will have the choice between an exchange with a higher execution probability for their clients (Exchange 1) and an exchange with a lower fee for them (Exchange 2). Compared to the Benchmark case where investors access markets directly and pay all exchange fees, there is a welfare loss when brokers route orders.

The conflict of interest faced by brokers creates a number of utility transfers. First, brokers earn positive profits from all investors who pay Commission c_H . However, fewer limit-order investors are willing to pay Commission c_H , to be routed to Exchange 1 in State $S = 2$, than the number who would be willing to pay the fee M_1 , if they could view the state and route their own order. These investors select Commission c_L instead, and are

¹³A full explanation of the Benchmark is located in the proof of Proposition 3.

always routed to Exchange 2. The utility of limit-order investors declines as a result of fewer limit-order investors realizing the private value y from their trades.

When more investors are routed to Exchange 2 in State $S = 2$, market maker, liquidity investor and informed trader profits also change. Market makers earn higher profits, as their orders execute more frequently at Exchange 2. The reverse is true for liquidity traders and informed traders, who pay higher fees at Exchange 2.

In aggregate, many of these transfers are offsetting. However, the loss of private value from limit-order investors, who move from trading at Exchange 1 to Exchange 2 in State $S = 2$, is not picked up by other agents. The result is a net loss in welfare compared to the Benchmark case. The total welfare change between the equilibrium presented in Proposition 1 and the benchmark case can be written as:

$$\int_{\bar{y}_{H,B}}^{\bar{y}_H} f(y)p_2(\theta_2 - \theta_1)y \cdot dy. \quad (19)$$

In Equation 19, the value $y_{H,B}$ represents the equilibrium private value such that investors select Exchange 1 in State $S = 2$ in the Benchmark, while y_H represents this equilibrium value with brokers. It can be shown that, $y_{H,B} \leq y_H$, and thus, this change is negative. Further, $y_{H,B}$ is a function of M_1 while y_H is a function of c_H , and thus, this welfare loss is increasing in the difference between c_H and M_1 .

Proposition 3 has implications for regulators. One way for regulators to remove the conflict of interest is by forcing exchange to charge identical liquidity-making fees. However, while the conflict of interest is removed, depending on the fee selected, welfare may not necessarily increase. Consider a regulator who mandates a uniform maker fee M , which is greater than the existing fee at Exchange 2 ($M > M_2$). If all exchanges are identical, brokers charge a single commission $\bar{c}$ and there is a single threshold private value $\bar{y}$. However,

compared to the equilibrium presented in Proposition 1, $\bar{y} > y_L$ since $M > M_2$. Thus, some investors with low private values will leave the market, offsetting some or all of the welfare gains from the high private value investors who are no longer impacted by the conflict of interest. This case applies equally to a regulator who simply prohibits rebates, as many incumbent exchanges offer relatively high rebates for making liquidity. Alternatively, in the simple setting of the model, mandating a single fee of $M = M_2$ allows regulators to remove the conflict of interest and obtain total welfare equal to the Benchmark in Proposition 3.

B. Comparative Statics

In this section, I present comparative static results related to broker and investor behaviour. First, I discuss changes in the equilibrium broker commissions. Second, I discuss changes in exchange selection from investors.

Proposition 4 (Equilibrium Broker Commissions):

- (i) The higher commission, c_H , is increasing in the difference between the liquidity making fees at Exchange 1 and Exchange 2 and in the probability of informed trading.*
- (ii) The lower commission, c_L , depends only on the liquidity making fee at Exchange 2.*

Proposition 4 contains the comparative statics related to commissions. Because of the incentive compatibility constraint, the difference between the high and low commission is much greater than difference in costs incurred by the broker. In equilibrium, brokers earn positive profits for the high commission, but charge only the marginal cost for the low commission. This comparative static result creates two testable implications based on fees at both exchanges as well as on adverse selection.

Testable Implication 2 (Commissions and Routing):

When brokers charge higher commissions, they will route some limit orders to venues with high maker fees. When brokers charge lower commissions, they will send all limit orders to venues with low maker fees.

Testable Implication 3 (Routing and the Probability of Informed Trading):

When the probability of informed trading is high, brokers will send more limit orders to venues with low maker fees than when the probability of informed trading is lower.

A large difference in fees creates a larger difference in costs to brokers, thus, to maintain incentive compatibility, they must be compensated. Practically speaking, this implies that discount brokers, who charge lower commissions, are more likely to send limit orders exclusively to low maker-fee venues, while full-service brokers likely to send some limit orders to venues with high maker fees.

Similarly, when the probability of informed trading is higher, brokers require additional compensation to route to Exchange 1. This is because the increase in the probability of informed trading causes the total probability of execution to converge across all exchanges. As the execution probabilities become more similar, rebates become more important and brokers are incentivized to concentrate their orders at exchanges with high maker rebates. In the model, brokers would adjust their commission if market conditions were changed. However, in practice, brokers are unlikely to adjust their commissions intraday, or across similarly-sized equities, making adjustments to routing behaviour more likely. Practically speaking, this is potentially testable for either market wide information events, or on a cross-sectional basis between more and less volatile assets. Higher volatility assets are more likely

to have all limit orders routed to maker-taker venues by brokers, while low volatility assets are more likely to have some orders sent inverse exchanges.

Proposition 5 (Limit-Order Investor Sorting):

- (i) *The private value y_H , is increasing in the difference between the liquidity-making fees at Exchange 1 and Exchange 2.*
- (ii) *The private value, y_L , is independent of the liquidity-making fee at Exchange 1.*

Proposition 5 relates the impact of fees on the order submission strategy of limit-order investors. Limit-order investors who wish to occasionally be routed to Exchange 1 are impacted by fees at both venues. The higher commission, c_H is increasing in the difference between fees at the two venues. As this commission increases, so too does the private value y_H , decreasing the total orders submitted at Commission c_H and increasing the orders submitted at Commission c_L .

IV. Alternative Commission Structures

In this model, the welfare-maximizing outcome (referred to earlier as the Benchmark case) occurs when investors view the state of the order book, pay exchange fees directly and route their own order flow. It is difficult to achieve this first-best outcome directly, as it would require investors to conduct expensive investment in services such as smart-order routers and colocation at multiple venues, the cost of which likely exceeds the welfare loss itself.

In this section, I present two alternative commission structures: brokers who pass exchange fees through to investors but control routing, and brokers who pass exchange fees through to investors who direct their own order flow. It has been argued that passed-through fees can

help alleviate the conflict of interest issues in maker-taker pricing regimes (Angel, Harris, and Spatt 2011). This model supports that finding with an important caveat: If fees are passed through to investors, but they do not have the ability to signal their state-contingent routing desires, the first best is not always obtainable.

A. Passed-Through Fees

I consider brokers who pass fees to investors but continue to control order routing. I model this through Assumption B1.

Assumption B1: Brokers pay no exchange fees, instead, these fees are paid directly by limit-order investors. Investors are unable to signal their state-contingent routing desire to brokers in any way.

When brokers pass fees through to investors, the equilibrium commission fees change such that there is a single equilibrium $c = 0$. In this case, all exchanges earn the broker zero profit in all states, and thus the broker is completely indifferent to all routing schemes. This presents a problem for investors: some may wish to be routed to Exchange 1 in State $S = 2$, while others may wish to be routed to Exchange 2.

Proposition 6 (Market Preference with Fee Pass-Through):

When exchange fees are passed on by the brokers, if $M_1 - M_2 \leq \frac{1-\delta}{1+\delta}\delta\sigma$, then $\bar{y}_2 \geq \bar{y}_1$ and all limit-order investors prefer being routed to Exchange 1 in State $S = 2$. Otherwise, there may be some investors who prefer to be routed to Exchange 2.

Proposition 6 shows the conditions under which some limit-order investors have different routing preference, even under pass-through fees. While the execution probability at Exchange 1 is higher than Exchange 2 in State $S = 2$, Exchange 1 also charges investors higher fees

for limit orders. If the difference in the expected value of the security is larger than the difference in exchange fees, all investors will have the same preference to be routed to the same exchange. Alternatively, if the difference in fees is large enough, it may dominate the difference in security value.

If brokers do not take state-contingent routing directions, or are unable to signal their state-contingent routing intentions when they earn zero profits at both exchanges, investors are unable to sort between exchanges as desired. If investors with a high private value y are routed to Exchange 2, rather than Exchange 1 in State $S = 2$, the execution probability is lower and they realize their private value less frequently. The loss of their private value y is not transferred to any other agent, and welfare declines. Similarly, if investors with a low private value $y = y_L$ are routed to Exchange 1, rather than Exchange 2 in State $S = 2$, they pay higher fees than expected. These investors earn negative profits in expectation, making them unwilling to submit orders and reducing welfare. Thus, when fees are passed through, but investors are unable to signal state-contingent routing intentions, the first-best welfare outcome is not always achievable.

B. Passed-Through Fees and Self-Directed Order Flow

A further refinement of passed-through fees is one in which investors select an exchange manually. I replace Assumption B1 with Assumption B2.

Assumption B2: Brokers pay no exchange fees, instead, these fees are paid directly by limit-order investors. Limit-order investors do not view the state of the order book, but each investor may direct their broker to route to a single exchange, regardless of the state.

Options similar to Assumption B2 are present on many popular brokerage platforms and give investors much more control over order handling. While investors on these platforms may

have access to updated prices and quotes, this data come with substantial latency compared to what is available to participants with smart-order routers and colocated servers. Thus, while investors are able to self-direct order flow, I assume that they are unable to view the true state of the order book.¹⁴

Proposition 7 (Self-Directed Order Flow):

- (i) Compared to the Benchmark, total welfare is lower if investors route their own orders and pay the associated exchange fee but do not view the state of the order book.*
- (ii) Compared to the case where brokers charge a commission and route orders, total welfare is higher if investors route their own orders and pay the associated exchange fee but do not view the state of the order book.*

Proposition 7 indicates that, when investors control their order flow, the first-best welfare outcome is not achievable. When investors cannot view the state, they must select a single exchange to trade at, regardless of the state. All investors would prefer to trade at Exchange 2 in State $S = 0$ and State $S = 1$. However, some investors may gain sufficient profits from trading at Exchange 1 in State $S = 2$, that they choose to trade at Exchange 1 in all states. These investors pay higher fees, and transfer profits to traders who use market orders and now pay lower fees. Costs for these investors at Exchange 1 rise, leading more investors to choose Exchange 2 in all states.

As in the case of Proposition 3, the investors who transfer to Exchange 2 realize fewer gains from trade in State $S = 2$. These gains are not transferred to other agents in the model, and welfare falls. However, compared to Proposition 3, the magnitude of welfare losses falls, suggesting that self-directed order flow may be preferable to conventional broker routing.

¹⁴A weaker assumption where investors view a noisy signal of the state ultimately leads to similar results.

V. Conclusion

The principal-agent relationship between brokers and their clients creates inefficiencies in financial markets. While existing theoretical literature has addressed the concept of exchange fees, a gap remains in explaining how these fees and rebates drive broker behavior and affect their clients. To explain these effects, I construct a model of limit-order trading in which brokers route limit orders from their clients.

I show that, with fixed price levels, rebates for making and taking liquidity drive broker routing decisions. Because brokers pay higher costs at exchanges with higher maker fees, they require compensation to route limit orders to these venues. This compensation comes in the form of a higher commission, which they charge their clients. When exchanges have similar fee structures, the commission which brokers charge is lower. Conversely, when adverse selection is higher, the commission brokers must charge is higher. The need for incentive compatibility creates a welfare loss: fewer investor orders are executed when they are sent to venues with low maker fees, lowering gains from trade.

This model offers a number of empirical predictions that remain available for testing. First, exchanges with lower liquidity-taking fees should see better market quality. An exchange with a lower taker fee should see a more retail market orders and, consequently, see less adverse selection and higher fill rates. Second, the broker's commission size dictates the broker's routing behavior. Brokers who charge a high commission should route orders to different venues, depending on market conditions, while those with low commissions will send all orders to low maker-fee venues. Finally, optimal limit order routing depends on the probability of informed trading. When the probability of informed trading is higher,

either market-wide or cross-sectionally between different assets, brokers should send more limit orders to maker-taker exchanges.

This model has key implications for regulators who wish to maximize welfare. Regulators can remove the broker's conflict of interest by forcing exchanges to charge identical maker fees. However, if this mandated maker fee is higher than existing maker fees, welfare is not necessarily increased. This suggests that prohibitions on rebates (specifically liquidity-making rebates) may mitigate the conflict of interest, but may not be the welfare-maximizing policy solution. An alternative policy which removes the broker's conflict of interest and reduces welfare losses is a mandate that brokers allow investors to self-direct order flow. Compared to the case when brokers route orders, investors who cannot view the order book directly suffer fewer losses, though they are still unable to achieve the welfare maximizing outcome.

REFERENCES

- Amihud, Yakov, and Haim Mendelson, 1980, Dealership market: Market-making with inventory, *Journal of Financial Economics* 8, 31–53.
- Anand, Amber, Tim McCormick, and Laura Serban, 2013, Incentives for liquidity provision: Is the make-take structure the answer?, *Working Paper*.
- Angel, James J, Lawrence E Harris, and Chester S Spatt, 2011, Equity trading in the 21st century, *The Quarterly Journal of Finance* 1, 1–53.
- Bakos, Yannis, Henry C Lucas Jr, Wonseok Oh, Gary Simon, Siva Viswanathan, and Bruce W Weber, 2005, The impact of e-commerce on competition in the retail brokerage industry, *Information Systems Research* 16, 352–371.
- Battalio, Robert, Shane A Corwin, and Robert Jennings, 2016, Can brokers have it all? On the relation between make-take fees and limit order execution quality, *The Journal of Finance* 71.
- Battalio, Robert, and Craig W Holden, 2001, A simple model of payment for order flow, internalization, and total trading cost, *Journal of Financial Markets* 4, 33–71.
- Boehmer, Ekkehart, Robert Jennings, and Li Wei, 2007, Public disclosure and private decisions: Equity market execution quality and order routing, *Review of Financial Studies* 20, 315–358.
- Brolley, Michael, and Katya Malinova, 2013, Informed trading and maker-taker fees in a low-latency limit order market, *SSRN Working Paper 2178102*.
- Chao, Yong, Chen Yao, and Mao Ye, 2015, Tick size constraints, two-sided markets, and competition between stock exchanges, *Working Paper* pp. Retrieved from: <http://www.economics.illinois.edu/seminars/documents/Ye.pdf>, October 28, 2016.
- Colliard, Jean-Edouard, and Thierry Foucault, 2012, Trading Fees and Efficiency in Limit Order Markets, *The Review of Financial Studies* 25, 3389–3421.
- Easley, David, and Maureen O’Hara, 1987, Price, trade size, and information in securities markets, *Journal of Financial Economics* 19, 69–90.
- Foucault, Thierry, 1999, Order flow composition and trading costs in a dynamic limit order market, *Journal of Financial Markets* 2, 99–134.
- Glosten, Lawrence R., 1994, Is the electronic open limit order boon inevitable?, *The Journal of Finance* 49, 1127–1161.

- , and Paul R. Milgrom, 1985, Bid, ask and transaction prices in a specialist market with heterogeneously informed traders, *Journal of Financial Economics* 14, 71–100.
- Ho, Thomas, and Hans R Stoll, 1981, Optimal dealer pricing under transactions and return uncertainty, *Journal of Financial economics* 9, 47–73.
- Maglaras, Costis, Ciamac C Moallemi, and Hua Zheng, 2015, Optimal execution in a limit order book and an associated microstructure market impact model, *SSRN Working Paper 2610808*.
- Malinova, Katya, and Andreas Park, 2015, Subsidizing liquidity: The impact of make/take fees on market quality, *The Journal of Finance* 70, 509–536.
- Milgrom, Paul, and Nancy Stokey, 1982, Information, trade and common knowledge, *Journal of Economic Theory* 26, 17–27.
- Parlour, Christine A, 1998, Price dynamics in limit order markets, *Review of Financial Studies* 11, 789–816.
- Parlour, Christine A., and Uday Rajan, 2003, Payment for order flow, *Journal of Financial Economics* 68, 379–411.
- Stoll, Hans R, 1978, The supply of dealer services in securities markets, *The Journal of Finance* 33, 1133–1151.
- Yim, Raymond, and Andrew Brzezinski, 2012, Limit order placement across multiple exchanges, *2012 IEEE Conference on Computational Intelligence for Financial Engineering and Economics (CIFEr)*.

A Appendix: Proofs

Proof of Lemma 1

From the perspective of a limit-order investor, the expected value of the asset at Exchange i in State S is given defined:

$$E[V|ex_i(S)] = \frac{-Pr(\text{Informed Trade}) \cdot \sigma}{\theta_i(S)} \quad (\text{A.20})$$

The probability of an informed trade at all exchanges and all states and is $Pr(\text{Informed Trade}) = \delta$. The total probability of order execution at Exchange i , $\theta_i(S)$, depends on the state of the order book.

Consider State $S = 0$, where no market maker posts. A limit-order investor receives all informed orders ($Pr = \delta$) and all uninformed orders ($Pr = 1 - \delta$) at either exchange. Thus:

$$\theta_1(S = 0) = \theta_2(S = 0) = 1, \quad (\text{A.21})$$

$$E[V|ex_1(S = 0)] = E[V|ex_2(S = 0)] = -\delta\sigma. \quad (\text{A.22})$$

Consider State $S = 1$, where a market maker posts at Exchange 1. An order routed to either exchange receives all informed orders ($Pr = \delta$), but only half of uninformed orders ($Pr = \frac{1}{2}(1 - \delta)$), as the others are absorbed by the market maker at Exchange 1. Thus,

$$\theta_1(S = 1) = \theta_2(S = 1) = \delta + \frac{1}{2}(1 - \delta), \quad (\text{A.23})$$

$$E[V|ex_1(S = 1)] = E[V|ex_2(S = 1)] = \frac{-\delta\sigma}{\delta + \frac{1}{2}(1 - \delta)}. \quad (\text{A.24})$$

Consider State $S = 2$, where a market maker posts at Exchange 2. An order routed to Exchange 1 receives all informed orders ($Pr = \delta$) and all uninformed orders ($Pr = 1 - \delta$). Thus:

$$\theta_1(S = 2) = 1, \quad (\text{A.25})$$

$$E[V|ex_1(S=2)] = -\delta\sigma. \quad (\text{A.26})$$

An order routed to Exchange 2 receives all informed orders ($Pr = \delta$), but only half of uninformed orders ($Pr = \frac{1}{2}(1 - \delta)$), as the others are absorbed by the market maker at Exchange 2. Thus,

$$\theta_2(S=2) = \delta + \frac{1}{2}(1 - \delta), \quad (\text{A.27})$$

$$E[V|ex_2(S=2)] = \frac{-\delta\sigma}{\delta + \frac{1}{2}(1 - \delta)}. \quad (\text{A.28})$$

Proof of Proposition 1

The equilibrium is weak perfect Bayesian equilibrium. Given Lemma 1, brokers and investors are aware of the execution probability and expected asset values at each exchange in each state. First, using a guess and verify approach, I determine the equilibrium broker commission schedule C . Second, given the equilibrium commissions and routing behaviour, limit-order investors sort based on the thresholds y_L and y_H .

Proof of Proposition 1: Commission Menus and Limit Order Routing

I conjecture that there exists an equilibrium in which all N brokers offer the same commission menu, consisting of 2 commissions, c_H and c_L . Define c_H as a commission under which brokers route limit orders to Exchange 1 in State $S = 2$ and to Exchange 2 in State $S = 0$ and State $S = 1$. Define c_L as a commission under which brokers route limit orders Exchange 2 in all states.

First, consider c_L , which I conjecture to be a zero-profit commission for Exchange 2:

$$c_L = M_2 \quad (\text{A.29})$$

Were the broker to route an order to Exchange 1 at this commission, he would earn a negative profit, given that $M_2 < M_1$. Thus, this commission is incentive compatible in all states. Any broker who deviated to offer a commission $c_L^* < c_L$ to route to Exchange 2 in all states

would earn negative profits. Any broker who deviated to offer a commission $c_L^* > c_L$ to route to Exchange 2 in all states would receive no share of investor orders, as $N - 1$ broker continue to offer c_L , and thus earn zero profits. Therefore, no deviation is profitable, and this commission can exist in equilibrium.

Second, consider c_H , which I conjecture to be the lower incentive compatible commission under which brokers route to Exchange 1 in State $S = 2$ and to Exchange 2 otherwise. This is defined as the lowest c_H such that:

$$\theta_1(S = 2)[c_H - M_1] \geq \theta_2(S = 2)[c_H - M_2] \quad (\text{A.30})$$

By inserting the values for $\theta_1(S = 2)$ and $\theta_2(S = 2)$, manipulation of Equation A.30 at equality results in:

$$c_H = \frac{2M_1 - (1 + \delta)M_2}{1 - \delta}. \quad (\text{A.31})$$

The Commission c_H is incentive compatible in State $S = 2$ by construction, and thus, brokers route to Exchange 1. In State $S = 0$ and State $S = 1$, Lemma 1 shows that the execution probabilities at each exchange are equal, and thus, it is incentive compatible to route to Exchange 2 since $M_2 < M_1$. Any broker who deviated to offer a commission $c_H^* < c_H$ would not find it incentive compatible to route to Exchange 1 in State $S = 2$, while any broker who offered $c_H^* > c_H$ would receive no share of investor orders. Thus, no deviation is profitable, and this commission can exist in equilibrium.

Finally, I conjecture that N brokers offering both c_L and c_H is an equilibrium menu. This claim requires two elements: (i) that it is not profitable for any broker to offer a third commission for some other routing strategy, and (ii) that if $N - 1$ brokers offer c_L and c_H it is not profitable for any one broker to not offer c_L or c_H .

A third commission would necessarily route to Exchange 1 in either of State $S = 0$ or State $S = 1$ to differentiate itself from the strategies in c_L and c_H . Consider a broker

charging some $\hat{c}$ who routes to Exchange 1 in either of State $S = 0$ or State $S = 1$. From Lemma 1, $\theta_1(S = 0) = \theta_2(S = 0)$ and $\theta_1(S = 1) = \theta_2(S = 1)$. Given that $M_2 < M_1$, there is no incentive compatible $\hat{c}$ such that a broker will route to Exchange 1 in either of State $S = 0$ or State $S = 1$. Thus, there can exist no such commission.

Consider $N - 1$ brokers offering c_L and c_H . A broker that did not offer c_H would lose the positive profits from the $\frac{1}{N}$ share of limit-order investors who would have selected this commission. The existing $N - 1$ brokers would acquire these clients at c_H and thus the deviating broker would gain no further profits at c_L . Thus, no single broker would not offer c_H . A broker that did not offer c_L would have no change in profits from the $\frac{1}{N}$ share of limit-order investors who would have selected this commission, as the commission is zero-profit. The existing $N - 1$ brokers would acquire these clients at c_L and thus the deviating broker would gain no further profits at c_H . Thus, no single broker has an incentive not to offer c_L . Given that there is no profitable deviation from this menu, it is a valid equilibrium.

Proof of Proposition 1: Limit-Order Investor's Problem

The expected utilities of an investor who selects commissions c_H and c_L are given by:

$$U_H(y) = p_1 \frac{(1 + \delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1 + \delta)} + y - c_H + \Delta \right] + p_2 [-\delta \cdot \sigma + y - c_H + \Delta] \\ + (1 - p_1 - p_2) [-\delta \cdot \sigma + y - c_H + \Delta], \quad (\text{A.32})$$

$$U_L(y) = p_1 \frac{(1 + \delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1 + \delta)} + y - c_L + \Delta \right] + p_2 \frac{(1 + \delta)}{2} \left[\frac{-2\delta \cdot \sigma}{(1 + \delta)} + y - c_L + \Delta \right] \\ + (1 - p_1 - p_2) [-\delta \cdot \sigma + y - c_L + \Delta]. \quad (\text{A.33})$$

Equation A.32 is greater than Equation A.33 for y sufficiently high. This threshold, y_H , is the value of y such that for all $y > y_H$ the limit-order investor prefers to select Commission c_H over the lower Commission c_L or not trading. Thus, y_H is defined as the lowest value y

such that:

$$U_H(y) \geq \max\{U_L(y), 0\}. \quad (\text{A.34})$$

Note that if $y_H > Y$, no investors are willing to select the high commission, and all investors choose either to trade at Commission c_L or not to trade.

Limit-order investors with a private value $y < y_H$ may choose either to pay the Commission c_L or not trade. Any investor who has a private value y such that Equation A.33 is positive is better off by trading at Commission c_L than not trading. Thus, the value y_L , such that investors with a value $y_H > y \geq y_L$ will choose to trade at Commission c_L , is defined as lowest value y that satisfies:

$$U_L(y) \geq 0 \quad (\text{A.35})$$

Note that if $y_L > y_H$, no investors are willing to select the Commission y_L , and all investors select to either pay the higher Commission c_H , or not trade.

Proof of Proposition 2

Proof follows directly from the equilibrium values computed in the Lemma 1.

Proof of Proposition 3

Consider the case where $Y \geq y_H > y_L$. The proof when this is not true follows an identical series of steps.

I construct a Benchmark equilibrium, where investors view the state of the order book, submit orders directly to exchanges, and pay the relevant exchange fee M_i instead of a broker commission. As before, all investors will prefer Exchange 2 in State $S = 0$ and State $S = 1$, as Lemma 1 shows market conditions are identical at both exchanges in those states. As was the case with brokers, some investors will prefer Exchange 1 in $S = 2$ ($y \geq y_H$), while others will prefer Exchange 2 ($y_H > y \geq y_L$) The equilibrium values y_H and y_L are computed in

similar manner to prior proofs, by simply replacing the relevant c_i values with M_1 and M_2 , depending on the State.

For the remainder of this proof, denote $y_{H,B}$ and $y_{L,B}$ as the private value threshold in the Benchmark without brokers and denote y_H and y_L as the thresholds in the case with brokers.

Consider State $S = 2$. Some investors prefer to be routed to Exchange 1, while other prefer Exchange 2. When comparing the benchmark case to the equilibrium with brokers, the value $y_L = y_{L,B}$ since $c_L = M_2$. Thus, no additional limit-order investors move from not-trading to trading at Exchange 2. As the same total number of limit orders are present, liquidity traders or informed traders execute the same quantity of orders.

However, when transitioning from the Benchmark to the equilibrium with brokers, the equilibrium value for $y_H > y_{H,B}$, since $c_H > M_1$. Thus, some investors who would have submitted their order to Exchange 1 in State $S = 2$ if they had controlled their own order flow will be unwilling to pay c_H . These investors will pay c_L and have their orders moved from Exchange 1 to Exchange 2 in $S = 2$.

The change in utility change for limit-order investors can be considered in two parts. First, a group of limit-order investors ceases trading at Exchange 1 in the Benchmark and begins trading at Exchange 2 with brokers. This change in utility can be given by:

$$\int_{y_{H,B}}^{y_H} f(y) p_2 [\theta_2(S = 2) [E[V|ex_2(S = 2)] + y + \Delta - c_L] - \theta_1(S = 2) [E[V|ex_1(S = 2)] + y + \Delta - M_1] dy. \quad (\text{A.36})$$

Second, a group of limit-order investors trade at Exchange 1 in both the Benchmark and the equilibrium with brokers. These limit-order investors begin paying the high Commission c_H ,

as opposed to the exchange fee M_1 . This change in utility can be characterized by:

$$\int_{y_H}^Y f(y) p_2 \theta_1(S=2) [M_1 - c_H] dy. \quad (\text{A.37})$$

These two utility changes are partially transferred to other agents in the model. Brokers earn increased profits from charging the incentive-compatible Commission c_H . Their profit, which exactly offsets Equation A.37, is given by:

$$\int_{y_H}^Y f(y) p_2 \theta_1(S=2) [c_H - M_1] dy. \quad (\text{A.38})$$

As brokers continue to earn zero profit from routing to Exchange 2 they have no other profit changes.

When limit-order investors transfer order from Exchange 1 to Exchange 2, market makers execute more orders at Exchange 2. Since the number of market orders is unchanged, their profit is directly in proportion to loss from limit-order investors. Given this behaviour, their change in expected profits from the Benchmark is given by:

$$\int_{y_{H,B}}^{y_H} f(y) p_2 \left[[-\delta\sigma + \Delta - M_2] - \frac{1}{2}(1 + \delta), \left[\frac{-\delta\sigma}{\frac{1}{2}(1 + \delta)} + \Delta - M_2 \right] \right] dy. \quad (\text{A.39})$$

Finally, informed traders and liquidity traders execute more orders at the higher cost Exchange 2, decreasing their respective profits by the following amounts:

$$\int_{y_{H,B}}^{y_H} f(y) p_2 \delta (T_1 - T_2) dy, \quad (\text{A.40})$$

$$\int_{y_{H,B}}^{y_H} f(y) p_2 (1 - \delta) (T_1 - T_2) dy. \quad (\text{A.41})$$

The total change in welfare can then be calculated by the sum of Equations A.36, A.37, A.38, A.39, A.40 and A.41. Through algebraic manipulation, and substituting $T_i = -M_i$,

this can be shown to be equal to:

$$\int_{y_{H,B}}^{y_H} f(y)p_2(\theta_2(S=2) - \theta_1(S=2))y \cdot dy. \quad (\text{A.42})$$

In equilibrium $\theta_2(S=2) < \theta_1(S=2)$ and $y_{H,BM} < y_H$. Thus, this integral is negative, representing a welfare loss.

In State $S=0$ and State $S=1$ the proof follows similar steps above. However, as $\theta_1(S=0) = \theta_2(S=0)$ and $\theta_1(S=1) = \theta_2(S=1)$, the welfare loss is equal to 0. Thus, welfare is lost only from State $S=2$.

Proof of Proposition 4

Commission c_H is given by:

$$c_H = \frac{2M_1 - (1 + \delta)M_2}{1 - \delta}. \quad (\text{A.43})$$

The value c_H is increasing in the liquidity making fee at Exchange 1, M_1 and decreasing in the liquidity making fee at Exchange 2, M_2 . Thus, an ex-ante increase in the difference between these values increases the equilibrium commission. From the same equation, c_H is increasing in δ under the conditions $0 < \delta < 1$, $M_1 > M_2$.

From Equation 12, c_L is given by:

$$c_L = M_2 \quad (\text{A.44})$$

This value is equal to the liquidity making fee at Exchange 2.

Proof of Proposition 5

From Equation 17, y_H is the lowest y such that:

$$U_H(y) \geq \max\{U_L(y), 0\}. \quad (\text{A.45})$$

Substituting the values from Equations 15 and 16 shows the value to be always dependent on c_H . The value of y such that Equation A.45 holds is increasing in c_H , which is itself increasing in the difference between the liquidity making exchange fees.

From Equation 18, y_L is the lowest y such that:

$$U_L(y) \geq 0 \quad (\text{A.46})$$

Substituting the value from Equation 16 shows the value to be dependent on c_L and independent of c_H . Since $c_L = M_2$, y_L is independent of the fee structure at Exchange 1.

Proof of Proposition 6

The exchanges have identical execution probabilities and expected asset values in states $S = 0$ and $S = 1$, and thus all investors prefer Exchange 2, as it has a lower fee.

Consider State $S = 2$. The total execution probability at Exchange 1 is higher than that at Exchange 2. Since Exchange 1 has a higher execution probability, all investors prefer Exchange 1 if the difference in the expected value of the asset is greater than the difference in fees:

$$E[V|ex_1(S = 2)] - E[V|ex_2(S = 2)] \geq M_1 - M_2 \quad (\text{A.47})$$

Substitution of the expected value equation and manipulation of this condition leads to the solution:

$$M_1 - M_2 \leq \frac{1 - \delta}{1 + \delta} \delta \sigma. \quad (\text{A.48})$$

Proof of Proposition 7

Consider the Benchmark case where $y_{H,B} > y_{L,B}$. Investors who submit to Exchange i , regardless of the State, have utility equal to:

$$U_i = \sum_S Pr(S) \theta_i(S) [E[V|ex_i(S)] + y - M_i + \Delta], \quad (\text{A.49})$$

Since $\theta_1(S = 0) = \theta_2(S = 0)$, $\theta_1(S = 1) = \theta_2(S = 1)$ and $\theta_1(S = 2) > \theta_2(S = 2)$, there exist y_1 and y_2 such that all investors with $y \geq y_1$ submit orders to Exchange 1, investors with $y_1 > y \geq y_2$ submit their order to Exchange 2 and investors with $\min\{y_1, y_2\} > y$ submit no order. I compare this equilibrium to the Benchmark, initially presented in Proposition 3.

First, consider the threshold y_2 , which results from submitting to Exchange 2 in all states. This threshold is identical to the threshold $y_{L,B}$ from the Benchmark. Thus, the same number of total limit orders are submitted, and there is no change in the number of liquidity traders or informed traders who execute orders.

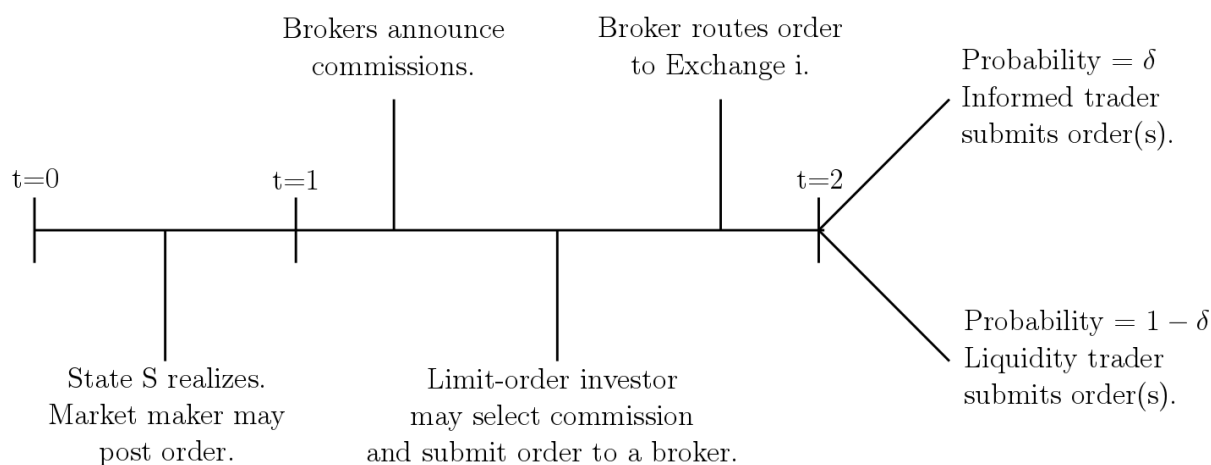
Second, consider the threshold y_1 , which results from submitting to Exchange 1 in all states. This threshold is higher than the Benchmark threshold, $y_{H,B}$, since investors are unable to route to Exchange 2 in State $S = 0$ and State $S = 1$, which would reduce their costs. Thus, some investors who would have selected to trade in Exchange 1 in all states will instead select to trade at Exchange 2 in all states. These investors realize fewer gains from trade.

The transfers to market makers, liquidity traders and informed traders are similar to the Proposition 3, with the only non-offsetting transfer being the investor's private value. The welfare loss is then equal to:

$$\int_{y_{H,B}}^{y_1} f(y) p_2(\theta_2(S = 2) - \theta_1(S = 2)) y \cdot dy. \quad (\text{A.50})$$

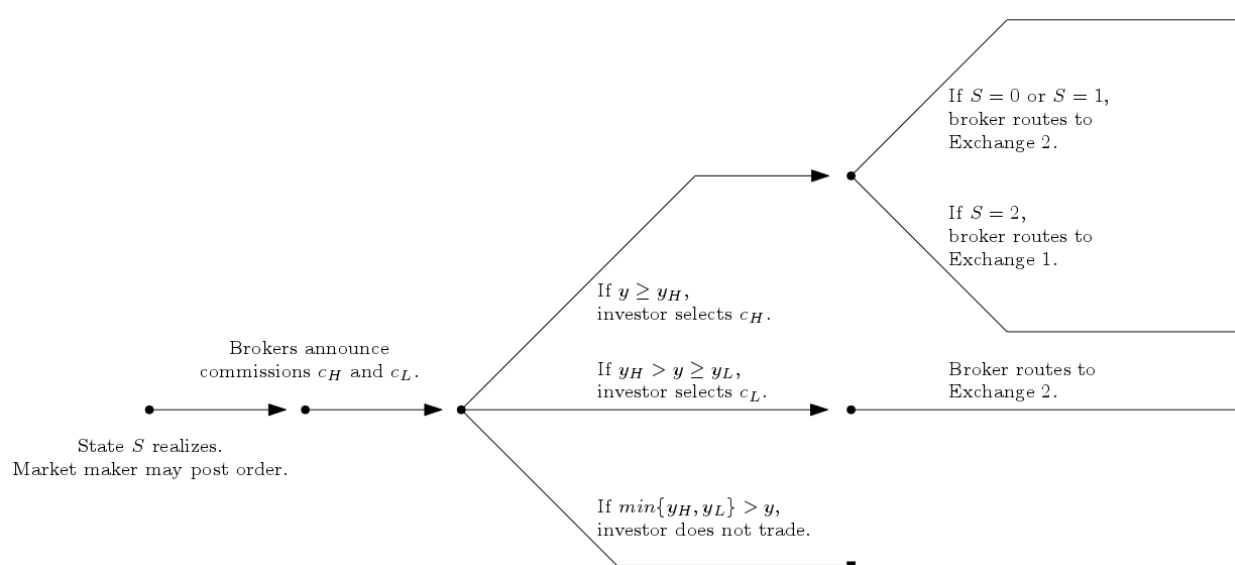
Since $M_1 < c_H$, $y_1 < y_H$ from the case when investors pay commissions. Thus, the magnitude of the loss is strictly lower in the case with self-directed orders than in the case where investors pay brokers to route their orders.

Figure 1
Model Timing



This figure illustrates the timing of the model. A market maker arrives in $t = 0$ and may submit an order, depending on the State S . In $t = 1$, brokers announce commission schedules. Investors view commissions and may submit an order, which their broker routes to an exchange. In $t = 2$, a liquidity demander arrives to submit a market order. With probability δ , the liquidity demander is an informed trader, while with probability $1 - \delta$, he is a liquidity trader.

Figure 2
Model Equilibrium



This figure illustrates the equilibrium actions for limit-order investors and their brokers. First, the State S is determined in $t = 0$, Second, all brokers announce identical commissions, c_H and c_L . Third, the limit-order investor will choose to either: (i) submit an order at c_H , (ii) submit an order at c_L , or (iii) not submit an order. She makes this decision based on her private value y . Finally, the broker will route the investor's orders. Orders that pay c_H will be routed to Exchange 2 in State $S = 0$ and State $S = 1$ and to Exchange 1 in State $S = 2$. Orders that pay c_L will be routed to Exchange 2 in all states.